

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“AlconGPT”

Submitted in partial fulfillment for the award of degree(20AI8CINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Samiksha Mehta - 1DS20AI050

Conducted at



Alcon

Department of Artificial Intelligence and Machine Learning

Dayananda Sagar College of Engineering Bangalore-78

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CERTIFICATE

This is to certify that the Internship titled “**AlconGPT**” carried out by **Ms SAMIKSHA MEHTA[1DS20AI050]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (**20AI8CINT**)

Signature of Reporting Head
Shweta Rawat
Team Lead
Alcon

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

Signature with Date

1) _____

2) _____

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **SAMIKSHA MEHTA**, final year student of Information Science and Engineering, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **ALCON**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:

USN: 1DS20AI050

Place: Bangalore

NAME: SAMIKSHA MEHTA

OFFER LETTER



Date: 22 January 2024

Dayananda Sagar College of Engineering

Shavige Malleswaram Hills, 91st Main Rd,
1st Stage, Kumaraswamy Layout,
Bengaluru, Karnataka 560078

AGREEMENT

SUB: Internship with Alcon Laboratories (India) Pvt. Ltd.

We would like to congratulate you on being selected for an internship with **Alcon** based at **Bangalore**.

Your internship is scheduled to start effective **5 Feb, 2024** for a period of **6 months** as per the internship request letter issued by **Dayananda Sagar College of Engineering** with letter dated **4 Dec 2023**.

You should report for the internship at the following address:

Alcon Laboratories (India) Pvt. Ltd.
Godrej Woodsman Estate RMZ Azure (IT Complex),
9th, 10th & 11th, Bellary Rd, Hebbal,
Bengaluru, Karnataka 560092

Reporting Time: 10:00 AM
Contact Person: Santhosh Kumar SG
Department: Human Resources (Talent Acquisition)

You will be eligible to receive a **stipend of INR 25,000 per month** during your internship. Other terms are mentioned in Annexure enclosed with this letter.

We look forward to welcome you and wishing you an enriching learning experience with us.

Sincerely,
Indumathy Karthikeyan
HR Head – ACS Bangalore, India

WE ARE
GREAT PLACE TO WORK - CERTIFIED™
Building and sustaining High-Trust, High-Performance Culture™



Acknowledgement:

I hereby acknowledge, confirm, and agree to the contents of this letter including the Annexure.

Samiksha Mehta

COMPLETION CERTIFICATE

A C K N O W L E D G E M E N T

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. B G Prasad, for providing us adequate facilities to undertake this Internship.

We would like to thank Dr. Vindhya P Malagi, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank Mr Gourav Terwardkar, Filoffee.AI, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, Prof. Kavya D N, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

SAMIKSHA MEHTA [1DS20AI050]

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CHAPTER 1

INTRODUCTION

In today's dynamic technological landscape, the intersection of innovation and user experience is paramount for companies striving to excel in their respective industries. This report encapsulates my ongoing internship experience at Alcon, a leading vision care company renowned for its commitment to enhancing vision and improving lives worldwide. Positioned within Alcon's esteemed IT department, I have been an integral member of the software engineering team, tasked primarily dealing with the UI/UX initiatives.

Throughout this internship, I have been immersed in the realm of software development, honing my skills and knowledge in crafting user-centric interfaces that seamlessly merge functionality with aesthetic appeal. Alcon's dedication to leveraging cutting-edge technology to revolutionize vision care has provided a fertile ground for exploration and growth within the realm of UI/UX design.

This report will delve into the intricacies of my role within Alcon's IT department, elucidating the scope of my work, the objectives set forth by the organization, and the challenges encountered along the journey. Moreover, it will reflect on the invaluable lessons learned, the achievements attained, and the recommendations formulated to further enhance the UI/UX endeavors at Alcon.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Alcon is the global leader in eye care, dedicated to helping people see brilliantly. With an over 75-year heritage, they are the largest eye care device company in the world, with complementary businesses in Surgical and Vision Care. Being a truly global company, they work in over 50 countries and serve patients in more than 140 countries. Alcon is grounded in cutting-edge innovation and breakthrough technology, transforming the way eye diseases and eye conditions are treated. They have the widest array of eye care offerings in the industry with products organized into two business: Surgical and Vision Care.

Surgical: Alcon offers the industry's most complete line of ophthalmic surgical products, enabling surgeons to achieve the best results for their patients. Their surgical portfolio includes technologies and devices for cataract, retinal, refractive surgery, as well as advanced technology intraocular lenses (ATIOLs) to treat cataracts and refractive errors, like presbyopia and astigmatism. They also provide advanced viscoelastics, surgical solutions, surgical packs, and other disposable products for cataract and vitreoretinal surgery.

Vision Care: Alcon is one of the largest manufacturers of contact lenses and lens care products. Their Vision Care portfolio offers a broad range of daily disposable, reusable and color-enhancing contact lenses and a comprehensive portfolio of ocular health products. These include products for dry eye, contact lens care and ocular allergies, as well as ocular vitamins and redness relievers.

CHAPTER 3

SCOPE OF WORK

During my internship at Alcon, my primary focus has been on the UI/UX aspect within the software engineering team. The scope of my work encompasses a range of responsibilities aimed at enhancing the usability, accessibility, and overall user experience of Alcon's digital products and platforms. Below are the key areas and tasks included in my scope of work:

- 1. User Interface Design:** Collaborating with cross-functional teams to conceptualize and design intuitive user interfaces for web and mobile applications. Creating wireframes, mockups, and prototypes to visualize design concepts and gather feedback from stakeholders. Implementing design principles and best practices to ensure consistency and coherence across all digital touchpoints.
- 2. User Experience Enhancement:** Conducting user research, including surveys, interviews, and usability testing, to gather insights into user needs and preferences. Analyzing user feedback and behavior data to identify pain points and opportunities for improvement in the user journey. Iteratively refining designs based on user feedback and usability testing results to optimize the overall user experience.
- 3. Accessibility Compliance:** Ensuring that digital products and interfaces adhere to accessibility standards and guidelines, such as WCAG (Web Content Accessibility Guidelines). Conducting accessibility audits and implementing remediation measures to enhance the inclusivity of digital experiences for users with disabilities.
- 4. UI/UX Documentation:** Documenting design decisions, rationale, and guidelines to maintain a centralized repository of UI/UX assets and resources. Creating style guides, pattern libraries, and design systems to facilitate consistency and scalability in UI/UX design across projects.

5. Collaboration and Communication: Engaging in regular meetings, stand-ups, and design reviews with cross-functional teams to ensure alignment and collaboration. Effectively communicating design ideas, rationale, and feedback to stakeholders, including product managers, developers, and business stakeholders.

6. Continuous Learning and Improvement: Staying updated on industry trends, emerging technologies, and best practices in UI/UX design through self-directed learning and professional development activities. Seeking feedback from peers, mentors, and supervisors to identify areas for growth and improvement in UI/UX skills and competencies.

The project that I have been working on involve AlconGPT, an internal AI-powered tool designed to streamline enterprise search and facilitate document and text translation into various languages. I was tasked with designing the User Guide for the application. This project encompassed a range of responsibilities aimed at effectively communicating the functionality, features, and usage guidelines of AlconGpt to end-users within the organization.

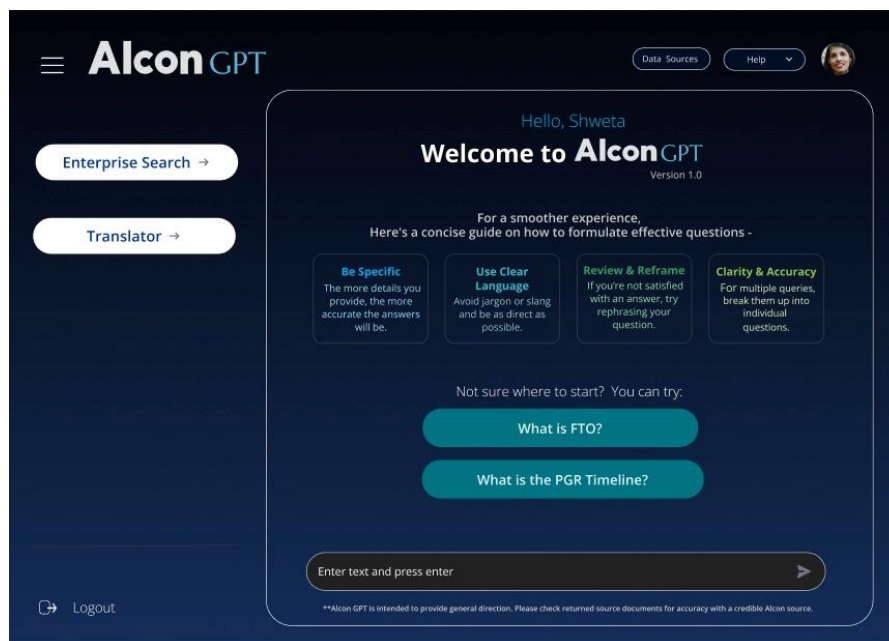


Fig 3.1

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

Throughout my internship at Alcon, the company has outlined specific learning objectives to guide my professional development and maximize the value of the internship experience. These objectives reflect Alcon's commitment to nurturing talent, fostering growth, and preparing interns for successful careers in the field of IT and software engineering. The following are the key learning objectives provided by Alcon:

Mastery of UI/UX Principles: Gain a deep understanding of user interface (UI) and user experience (UX) design principles, methodologies, and best practices. Learn to apply design thinking techniques to solve complex UI/UX challenges and create intuitive digital experiences for users.

Technical Proficiency in Design Tools and Technologies: Acquire proficiency in industry-standard design tools, such as Figma, to create high-fidelity UI mockups and prototypes. Develop skills in API development and Agile and Jira, to implement and iterate on UI designs effectively.

REST API Development: Acquire knowledge and proficiency in developing REST API using python, Flask and SQLAlchemy to aid in developing applications.

User-Centric Design Approach: Understand the importance of user research and usability testing in informing design decisions and validating design solutions. Learn to empathize with users, identify their needs and pain points, and incorporate user feedback into the design iteration process.

Collaboration and Communication Skills: Enhance communication and collaboration skills by effectively articulating design ideas, presenting design concepts, and soliciting feedback from cross-functional teams. Learn to work collaboratively with product managers, developers, and other stakeholders to translate business requirements into user-centric design solutions.

Adherence to Industry Standards and Guidelines: Familiarize oneself with industry standards and guidelines for UI/UX design, including accessibility standards (e.g., WCAG) and platform-specific design guidelines (e.g., Material Design, iOS Human Interface Guidelines). Develop the ability to ensure compliance with relevant standards and guidelines in UI/UX design projects.

Professional Growth and Adaptability: Cultivate a growth mindset and a willingness to embrace new challenges, feedback, and opportunities for learning and improvement. Adapt to a dynamic and fast-paced work environment, demonstrating resilience, flexibility, and a proactive approach to problem-solving.

By aligning my learning objectives with the strategic goals and priorities of Alcon, I aim to leverage this internship opportunity to acquire valuable skills, knowledge, and experiences that will contribute to my professional development and future success in the field of IT and software engineering.

CHAPTER 5

CHALLENGES AND SOLUTIONS

1. Scope Creep:

Challenge: Scope creep occurs when additional features or requirements are added to the project beyond the original scope, leading to timeline delays and resource overruns.

Solution: Implementing robust project management practices such as setting clear project scope boundaries, conducting regular scope reviews, and obtaining stakeholder sign-off on any scope changes. Agile methodologies like Scrum or Kanban can help manage changes effectively through iterative development cycles.

2. Content Organization and Prioritization:

Challenge: Determining the most effective way to organize and prioritize large amounts of information within a project, while maintaining comprehensiveness and conciseness, can be daunting.

Solution: Conducting thorough research to understand user needs and usage scenarios, which informs the creation of a structured information architecture that prioritizes key features and common user tasks. Regular feedback sessions with stakeholders help refine content organization and ensure alignment with user expectations.

3. Technology Dependencies and Integration Challenges:

Challenge: Projects often rely on external technologies, APIs, or third-party services, leading to dependencies and integration challenges that can affect project timelines and functionality.

Solution: Conducting thorough research and feasibility studies to identify potential technology dependencies early in the project lifecycle. Establishing clear communication channels and collaboration agreements with technology providers can help mitigate integration risks, while robust testing and fallback mechanisms can minimize the impact of integration failures.

4. User Adoption and Change Management:

Challenge: Successfully introducing new software or features to end-users and ensuring their adoption can be challenging, particularly when users are resistant to change or unfamiliar with the new tools.

Solution: Implementing effective change management strategies, including user training and onboarding programs, communication plans, and user feedback mechanisms. Involving end-users in the design and development process through user testing and co-creation can increase buy-in and acceptance of the new software.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

Throughout my internship at Alcon, I had the opportunity to make significant contributions to the software engineering team and the overall success of the AlconGpt project. The following highlights some of the key achievements and contributions made during the internship:

Development of the AlconGpt User Guide:

Designed and authored a comprehensive user guide for AlconGpt, detailing its features, functionalities, and usage guidelines.

Collaborated with stakeholders and subject matter experts to gather requirements and ensure the accuracy and relevance of the user guide content.

Implemented a structured information architecture and clear, user-friendly language to facilitate understanding and usability for end-users.

Enhancement of User Experience for AlconGpt:

Conducted usability testing and gathered user feedback to identify pain points and opportunities for improvement in the user interface and overall user experience of AlconGpt.

Proposed and implemented iterative design updates and enhancements based on user feedback, resulting in improved usability, efficiency, and user satisfaction.

Advocated for user-centric design principles and best practices within the software engineering team, promoting a culture of continuous improvement and customer focus.

Cross-Functional Collaboration and Communication:

Engaged in regular communication and collaboration with cross-functional teams, including product managers, developers, designers, and stakeholders, to align project goals and priorities.

Presented project updates, findings, and recommendations to senior leadership and key stakeholders, demonstrating professionalism, clarity, and confidence in communication.

Fostered a culture of teamwork and knowledge sharing within the software engineering team, actively participating in team meetings, brainstorming sessions, and peer reviews.

Continuous Learning and Professional Development:

Actively pursued opportunities for learning and skill development, including self-directed study, online courses, and participation in company-sponsored training programs.

Demonstrated adaptability and resilience in tackling new challenges and learning new technologies and methodologies relevant to software engineering and UI/UX design.

Received positive feedback and recognition from supervisors, colleagues, and stakeholders for dedication, initiative, and contributions to project success.

By leveraging these achievements and contributions, I have not only enhanced my skills and knowledge in software engineering and UI/UX design but also made tangible contributions to the success of the AlconGpt project and the broader goals of the software engineering team at Alcon.

CHAPTER 7

REFLECTION AND ANALYSIS

My internship experience at Alcon has been both enriching and insightful, providing me with valuable opportunities for growth, learning, and self-discovery. As I reflect on my journey and analyze the various aspects of my internship, several key insights and reflections emerge:

Professional Growth and Development:

Reflecting on my initial expectations and aspirations at the beginning of the internship, I am pleased to acknowledge the significant growth and development I have experienced over the course of my tenure at Alcon. Through hands-on experience, exposure to real-world projects, and mentorship from seasoned professionals, I have honed my technical skills, expanded my knowledge base, and gained a deeper understanding of software engineering and UI/UX design principles. Embracing new challenges, navigating complex projects, and overcoming obstacles have not only enhanced my confidence and competence but also instilled in me a sense of resilience and adaptability crucial for success in the dynamic field of technology.

Project Impact and Contribution:

Upon reflection, I am gratified by the meaningful contributions I have made to the AlconGpt project and the software engineering team at Alcon. Developing the user guide for AlconGpt, implementing usability enhancements, and advocating for accessibility and localization initiatives have not only improved the usability and accessibility of the tool but have also fostered a culture of user-centric design and continuous improvement within the team. My contributions have not only demonstrated my technical proficiency and problem-solving skills but have also reinforced the importance of collaboration, communication, and empathy in driving project success.

Lessons Learned and Areas for Improvement:

Reflecting on the challenges encountered and the lessons learned during my internship, I recognize the importance of embracing ambiguity, seeking feedback, and learning from failures.

I have learned the value of effective time management, prioritization, and iteration in project execution, as well as the importance of proactive communication and collaboration in cross-functional teams. Moving forward, I am committed to further enhancing my skills, deepening my knowledge, and refining my processes to become a more effective and impactful contributor in future endeavors.

Personal and Professional Growth:

Beyond the technical skills and project accomplishments, my internship at Alcon has also been a journey of personal growth and self-discovery. Engaging with diverse perspectives, collaborating with multidisciplinary teams, and navigating real-world challenges have broadened my horizons, expanded my worldview, and enriched my understanding of the intersection between technology and human experience. I am grateful for the mentorship, support, and camaraderie of my colleagues at Alcon, whose guidance and encouragement have empowered me to strive for excellence and pursue my passions with purpose and enthusiasm.

CHAPTER 8

RECOMMENDATIONS

Given the open and approachable nature of leaders and mentors at Alcon, along with the fostering and supportive company culture, the internship program can be further enhanced by implementing the following recommendations:

Facilitate Cross-Departmental Collaboration:

Create opportunities for interns to collaborate across different departments and teams within the organization. Encourage leaders and mentors to facilitate introductions and connections, allowing interns to gain insights into various aspects of the business and expand their network.

Promote Peer Learning and Knowledge Sharing:

Foster a culture of peer learning and knowledge sharing among interns by organizing regular meetups, brainstorming sessions, or knowledge-sharing events. Encourage interns to share their experiences, insights, and skills with each other, creating a supportive community of learners.

Offer Flexibility in Project Assignments:

Provide interns with flexibility in choosing or customizing their project assignments based on their interests, skills, and career goals. Empower them to pursue projects that align with their passions and aspirations, fostering a sense of ownership and investment in their work.

By implementing these recommendations, Alcon can further enrich the internship experience, empowering interns to learn, grow, and thrive in a supportive and inclusive environment while making meaningful contributions to the organization's success.

CHAPTER 9

CONCLUSION

My internship experience at Alcon has been nothing short of transformative, providing me with invaluable opportunities for learning, growth, and personal development. Throughout my tenure, I have had the privilege of working alongside dedicated professionals who embody the organization's values of openness, approachability, and fostering a supportive company culture.

During my time at Alcon, I had the opportunity to contribute to the development of the AlconGpt project, particularly in the creation of the user guide, as well as in enhancing user experience and accessibility. These experiences have not only strengthened my technical skills but have also deepened my understanding of the importance of user-centric design and cross-functional collaboration in delivering impactful solutions.

Moreover, the mentorship and guidance I received from leaders and mentors at Alcon have been instrumental in shaping my professional journey. Their openness, approachability, and willingness to share their knowledge and insights have been invaluable assets, fostering an environment where I felt supported and empowered to take on new challenges and pursue my passions.

As I conclude my internship at Alcon, I carry with me a profound sense of gratitude for the experiences, lessons, and relationships forged during my time here. I am confident that the skills, knowledge, and experiences gained will serve as a strong foundation for my future endeavors in the field of software engineering and UI/UX design.

I extend my heartfelt thanks to everyone at Alcon who has contributed to making my internship experience truly exceptional. I am grateful for the opportunity to have been part of such a dynamic and innovative organization and look forward to applying the lessons learned and making a positive impact in my future endeavors.

CHAPTER 10

REFERENCES

Name: Shweta Rawat

Position: Sr. UI/UX Designer

Company: Alcon

Contact: shweta.rawat@alcon.com